

FINAL EXPLANATION: GENERAL SCIENCE GRADE 10

Teacher Key: Exemplar Answers and Marking Guide

FINAL EXPLANATION: GENERAL SCIENCE GRADE 10: THE PERIODIC TABLE (SUB-STRAND 2.1)

Teacher Key: Exemplar Answers (do not give to students)

INSTRUCTIONS GIVEN TO STUDENTS

Final Explanation: Why do fertiliser bags carry such specific formulae? Think about the two bags a Grade 10 learner in Kericho saw on her family's tea farm: CAN, $\text{Ca}(\text{NO}_3)_2 \cdot \text{NH}_4\text{NO}_3$, and DAP, $(\text{NH}_4)_2\text{HPO}_4$. Both are white crystalline solids that dissolve in water, and both contain nitrogen, yet their formulae are different. Answer each prompt in the blank space under it, using what you learned in Lessons 1 to 8. Write in full sentences. Where a question asks you to show working, write each step. You may draw dot-and-cross diagrams or small tables. Use the words you met in class (valence electrons, group, period, ion, valency, radical, charge balance) and link each answer back to the fertiliser bags.

Part 1: How the periodic table is organised

A learner in Kericho says: 'The periodic table is just a list of elements. It has nothing to do with the fertiliser on our tea farm.' Use calcium (found in CAN) and nitrogen (found in CAN and DAP) to show that she is wrong. In your answer: (a) explain what the periodic law is and how you discovered it when you sorted the 20 element cards; (b) explain what the periods and the groups of the periodic table tell you about an atom's electron arrangement; (c) state which group calcium and nitrogen are in (use the group numbers 1 to 8 for the main groups, as in Lessons 4 and 5) and give the reason for each.

(a) When we sorted the 20 element cards in order of atomic number and looked at the number of electrons in the outermost shell (valence electrons), the numbers repeated in a pattern: 1, 2, 3 and so on up to 8 (the noble gases), and then the pattern started again. This repeating pattern is the periodic law. (b) Rows are called periods, and the period number tells us how many electron shells are occupied. Columns are called groups, and for the main groups numbered 1 to 8 the group number tells us the number of valence electrons, so elements in the same group have similar outer-shell arrangements. (c) Calcium (electron arrangement 2,8,8,2) has 2 valence electrons, so it is in Group 2. Nitrogen (2,5) has 5 valence electrons, so it is in Group 5. The periodic table therefore organises elements by the same electron arrangement that decides how they combine, so it is connected to the formulae on the CAN and DAP bags.

Part 2: Why electron arrangement is significant

Explain why the arrangement of electrons in an atom, especially the outer shell, decides what ion the atom forms. Use calcium (2,8,8,2) and nitrogen (2,5) as your two examples. In your answer: (a) say what a full outer shell means for the stability of an atom and name one noble gas example; (b) explain, with the electron changes, how calcium forms its ion and how nitrogen forms its ion, giving each ion's symbol and charge; (c) explain why calcium forms a 2+ ion and not a 1+ or 3+ ion, using the noble gas rule.	(a) An atom is most stable when its outer shell is full, which is the electron arrangement of a noble gas. For example, neon (2,8) and argon (2,8,8) have full outer shells and hardly react. (b) Calcium (2,8,8,2) has 2 outer electrons. It loses these 2 electrons and is left with 2,8,8, the arrangement of argon. It now has 2 more protons than electrons, so it forms the cation Ca^{2+}. Nitrogen (2,5) has 5 outer electrons and is 3 electrons short of a full outer shell. It gains 3 electrons to reach 2,8, the arrangement of neon, so it forms the anion N^{3-}. (c) Calcium reaches a noble gas arrangement (argon, 2,8,8) only by losing exactly 2 electrons. If it lost only 1 electron it would be 2,8,8,1, which is not a noble gas arrangement because it still has a single electron in its outer shell. Losing a 3rd electron would mean removing an electron from the full argon-like shell (2,8,8) that has just been reached. That shell is already stable, so it holds its electrons strongly and removing one would need far too much energy, so it does not happen. So the charge of calcium's ion is fixed by its electron arrangement and is always 2+.
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Part 3: From ion charge to the formula $\text{Ca}(\text{NO}_3)_2$

Use the idea of valency and the criss-cross rule to write the formula of calcium nitrate. In your answer: (a) state the charge of the calcium ion and of the nitrate radical, and say what valency means; (b) show every step of the criss-cross method, writing the subscript that each symbol receives; (c) say how you decided whether the NO_3 part of your formula needs brackets; (d) check your formula by showing the total positive charge and the total negative charge.	(a) The calcium ion is Ca^{2+} and the nitrate radical is NO_3^-. Valency is the combining power of an element or radical, and it equals the size of the ion charge. So calcium has valency 2 and nitrate has valency 1. (b) Step 1: write the symbols with their valencies: Ca (2) and NO_3 (1). Step 2: criss-cross the numbers: the valency of nitrate, 1, becomes the subscript of Ca, giving Ca_1, and the valency of calcium, 2, becomes the subscript of NO_3, giving NO_3 with a subscript of 2. Step 3: a subscript of 1 is not written, so Ca_1 is written Ca. Because NO_3 is a radical and its subscript is 2, the whole radical is put in brackets, giving $\text{Ca}(\text{NO}_3)_2$. Step 4: verify the formula: $\text{Ca}(\text{NO}_3)_2$ contains 1 Ca, 2 N and 6 O, and the charges balance (see part d). (c) NO_3 is a radical, a group of atoms that stays together as one unit. When more than one unit is needed, the subscript must apply to the whole unit (one N and three O), so the radical is put in brackets before the subscript is written. (d) Total positive charge: $1 \times (2+) = 2+$. Total negative charge: $2 \times (1-) = 2-$. They cancel, so the compound is electrically neutral.
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Part 4: Why CAN and DAP have different formulae

Both fertiliser bags contain nitrogen, but the formulae on the two bags are different. Look at the NH_4NO_3 part of the CAN	(a) NH_4^+ has a charge of 1+ and NO_3^- has a charge of 1-. One of each gives 1+ and 1-, which cancel, so the ratio is 1 : 1 and the formula is NH_4NO_3. (b) In $(\text{NH}_4)_2\text{HPO}_4$ there are two NH_4^+ ions, so the total positive charge is $2 \times (1+) = 2+$. For the compound to be neutral, the remaining part of the formula, HPO_4, must carry a total negative charge of 2-. (c) In both formulae the numbers of ions are chosen so that total positive charge equals total negative charge. The formulae differ because the negative part of each
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formula and at the DAP formula, $(\text{NH}_4)_2\text{HPO}_4$. In your answer: (a) give the charge of NH_4^+ and of NO_3^- and decide what ratio of these two ions gives an electrically neutral compound; (b) count the NH_4^+ ions in $(\text{NH}_4)_2\text{HPO}_4$ and work out the total positive charge they carry; (c) explain in one or two sentences what the two formulae have in common and why they still look different.	compound carries a different total charge (1– for nitrate, 2– for HPO_4), so a different number of ammonium ions is needed to balance it. This is charge balance, and it comes from the valency of each ion, which comes from electron arrangement.
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Part 5: Answering the driving question

Driving question: Why do the fertiliser bags on Kenyan farms carry such specific and different chemical formulae, and what does the inner structure of an atom have to do with it? Write a clear explanation for the learner in Kericho. Your answer must: (a) link the inner structure of the atom to the formula on the bag, putting the steps of your reasoning in the correct order; (b) use calcium and the CAN formula as your worked example; (c) use the balanced equation $\text{Na} + \text{Cl}_2 \rightarrow \text{NaCl}$ or $\text{Mg} + \text{O}_2 \rightarrow \text{MgO}$ from Lesson 7 to show what balancing does and why only coefficients, not subscripts, can be changed; (d) say in one sentence what you still wonder about.	(a) The inner structure of an atom, specifically the number of electrons in its outer shell, decides how many electrons the atom loses, gains or shares in order to reach a stable full outer shell. This gives each ion a fixed charge, and the size of that charge is its valency. Because every compound must be electrically neutral, ions combine in the only ratio where total positive charge equals total negative charge. Cross-multiplying the valencies gives that ratio, and it is written as the formula on the bag. (b) Calcium (2,8,8,2) is in Group 2. It loses 2 electrons to form Ca^{2+} with valency 2. Nitrate is NO_3^- with valency 1. Criss-crossing gives $\text{Ca}(\text{NO}_3)_2$, so each calcium ion needs two nitrate ions to balance its 2+ charge. This is why the farmer in Kericho sees $\text{Ca}(\text{NO}_3)_2$ and not CaNO_3. The NH_4NO_3 part of CAN is one NH_4^+ balancing one NO_3^-. (c) Take $\text{Mg} + \text{O}_2 \rightarrow \text{MgO}$. Unbalanced, there are 2 O atoms on the left and 1 on the right. The formula MgO is fixed by the valencies (Mg^{2+} and O^{2-}), so we cannot change its subscripts. Instead we write 2 as a coefficient: $2\text{Mg} + \text{O}_2 \rightarrow 2\text{MgO}$. Now there are 2 Mg and 2 O atoms on each side, so no atoms are created or destroyed. Changing a subscript, for example writing MgO_2, would make a different substance, whereas a coefficient only changes how many units of the correct formula take part. The same rule applies to formulae like $\text{Ca}(\text{NO}_3)_2$ and $(\text{NH}_4)_2\text{HPO}_4$: their subscripts come from charge balance and must stay as they are. (d) Example answer: I still wonder why plants respond differently to CAN and DAP at different stages of growth, since both contain nitrogen.
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Criterion	Excellent (4)	Proficient (3)	Developing (1–2)
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Linking atomic structure to the periodic table	Clearly explains the periodic law, periods and groups, and correctly links calcium and nitrogen to their groups through their valence electrons.	Explains most ideas about periods, groups and valence electrons, with one small gap or error in placing calcium or nitrogen.	Describes the periodic table only as a list, or does not connect groups to valence electrons.
Explaining ion formation	Correctly shows electron loss or gain for Ca and N, gives Ca^{2+} and N^{3-} , and explains the noble-gas stability reason, including why the charge is fixed.	Gives correct ions and electron changes but the reasoning about stability or fixed charge is incomplete.	States ions without explaining the electron changes, or confuses cations and anions.
Writing and checking formulae	Applies the criss-cross rule accurately, uses brackets correctly for radicals, and verifies charge balance with numbers for $\text{Ca}(\text{NO}_3)_2$, NH_4NO_3 and $(\text{NH}_4)_2\text{HPO}_4$.	Writes correct formulae with the criss-cross rule but gives limited checking of charge balance or explanation of brackets.	Writes formulae incorrectly or without the steps, and does not check that charges balance.
Connecting the whole explanation to the phenomenon	Builds a complete, ordered causal chain from electron arrangement to valency to formula on the CAN and DAP bags, uses a balanced equation correctly to show conservation of atoms and why only coefficients change, and refers throughout to the Kericho learner.	Gives a mostly correct chain with one link missing or weakly explained, and refers to the fertiliser bags.	Gives disconnected statements and does not link atomic structure to the formulae on the bags.

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